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Deep Reinforcement Learning-based Energy Management and Fault-Tolerant Control for Vehicle-to-Grid Enabled Electric Vehicle Drive Systems

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***Abstract*—Vehicle-to-Grid (V2G) technology has emerged as an effective approach for supporting bidirectional energy exchange between electric vehicles and smart grids. However, uncertainties associated with renewable energy generation, electricity price fluctuations, varying driving conditions, and component faults make real-time energy management and reliable system operation challenging. This study presents a Deep Reinforcement Learning (DRL)-based framework that integrates intelligent energy management with adaptive fault-tolerant control for V2G-enabled electric vehicle drive systems. A Deep Q-Network (DQN) agent continuously evaluates battery state-of-charge, renewable power availability, load demand, electricity pricing, and fault severity to determine optimal charging, discharging, and grid-support actions while maintaining stable system operation under abnormal conditions. The proposed framework is validated using MATLAB/Simulink under diverse driving cycles, renewable energy scenarios, and component fault conditions. Simulation results demonstrate an energy efficiency of 96.1%, battery utilization of 94.2%, grid support capability of 91.3%, and a 34.7% reduction in operating cost compared with a conventional rule-based strategy. The integrated fault diagnosis and adaptive recovery mechanism achieves 98.8% fault detection accuracy, identifies faults within 24 ms, and restores normal operation in 0.14 s. These results demonstrate the effectiveness of the proposed framework in improving the efficiency, reliability, and operational resilience of next-generation V2G-enabled electric vehicle drive systems.**

Keywords— Electric Vehicle (EV), Vehicle-to-Grid (V2G), Deep Reinforcement Learning (DRL), Deep Q-Network (DQN), Energy Management System, Fault-Tolerant Control, Smart Grid, Battery Management, Fault Diagnosis.

I. INTRODUCTION

The growing popularity of electric vehicles (EVs), advancements in battery technology, and the growing demand for sustainable energy solutions are driving a substantial transformation in the transportation sector. Globally, governments and businesses are collaborating to cut carbon emissions. They accomplish this by increasing the production of electric cars (EV). Electric vehicles (EVs) can be a viable alternative to traditional gasoline-powered vehicles since they are more energy efficient, emit fewer pollutants, and use less fossil fuels. The large-scale integration of electric vehicles into current power systems raises a variety of technological problems, including energy management, grid stability, operational reliability, and charging infrastructure [1].

Vehicle-to-Grid (V2G) technology has sparked a lot of attention as a potential solution to a variety of concerns. Unlike traditional charging systems, which only allow energy to flow from the grid to the vehicle, V2G systems enable power to flow in both directions between the utility grid and the electric car batteries. Electric vehicles can function as distributed energy storage systems because to their intrinsic properties, which allow for the integration of renewable energy sources, frequency management, voltage stabilisation, and peak load demand. When power demand is low, EV batteries can be charged at a lesser cost, and when demand is high, stored energy can be released back into the grid. This two-way exchange maximises the utilisation of renewable energy resources, improves system flexibility, and generates additional revenue.

Despite the advantages that V2G technology provides, the implementation of adequate energy management systems remains a significant hurdle. Variations in driving behaviour, battery state-of-charge (SOC), power price, renewable energy availability, and grid operating circumstances are the primary factors influencing the extremely dynamic operating environment of V2G-enabled electric vehicle systems. Traditional energy management solutions, such as rule-based controllers and optimisation algorithms, are frequently based on pre-established operating conditions and mathematical assumptions that may be insufficient to deal with real-world unpredictability [3]. These methods have the potential to reduce energy efficiency, increase battery deterioration, provide inadequate charging and discharging options, and reduce adaptability to changing system conditions.

Recent advances in machine learning and artificial intelligence have opened new opportunities for intelligent energy management in electric vehicle applications. Deep Reinforcement Learning (DRL) is widely recognised as a sophisticated decision-making system that constantly interacts with complicated environments to develop the optimum control techniques. Unlike supervised learning approaches, which require huge, labelled datasets, DRL agents learn directly from their environment by maximising cumulative

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rewards. This feature makes the DRL approach ideal for energy management problems involving multiple objectives, uncertain operating conditions, and nonlinear system dynamics [4]. By continuously monitoring system variables such as battery charge, load demand, renewable energy generation, and electricity pricing signals, DRL agents can improve overall system performance by learning adaptive charging and discharging techniques.

Reliability and fault tolerance are essential considerations for V2G-enabled electric vehicle propulsion systems, alongside energy management. The system components include reversible DC–DC converters, voltage source inverters, electric traction motors, sensors, communication interfaces, grid-connected power electronic converters, and lithium-ion batteries. Malfunction of any of these components can substantially impair the system's operation, leading to diminished efficiency, power interruptions, equipment damage, or safety risks. The predominant faults are communication failures, malfunctioning converter switches, inoperative batteries, and defective motor winding [5]. Thus, the advancement of fault-tolerant control systems and reliable defect identification tools has become a vital research domain to guarantee the consistent and dependable functioning of systems.

Energy management systems are generally incompatible with traditional fault-tolerant control methodologies. Numerous studies classify energy optimization, fault recovery, and fault identification as separate control issues. This hampers the attainment of both energy efficiency and system reliability. Although these solutions may effectively address difficulties, they may not represent the most efficient use of energy when systems are malfunctioning. Moreover, most contemporary energy management systems fail to explicitly integrate fault severity data into their decision-making processes, diminishing their efficacy during unforeseen breakdowns.

This study introduces a cohesive framework for energy management and resilient control of electric vehicle drive systems capable of grid connectivity. It is founded on Deep Reinforcement Learning (DRL). This would eliminate the previously identified deficiencies. The proposed framework amalgamates battery state-of-charge, renewable energy accessibility, electricity tariffs, load requirements, converter operational parameters, and fault intensity into a unified decision-making process, unlike conventional approaches that optimize energy management and fault recovery separately. The integrated fault diagnosis module enables the reconfiguration of adaptive control in atypical operating conditions. A Deep Q-Network (DQN) agent perpetually acquires optimal methods for charging, discharging, and providing grid support. The suggested framework functions as a comprehensive control method for intelligent V2G-enabled electric vehicle applications by simultaneously achieving energy economy, operational reliability, battery longevity, and utility assistance.

The paper presents three essential contributions. Initially, a structure for synchronized energy administration and resilient control of grid-integrated electric vehicle propulsion systems is established utilizing Deep Reinforcement Learning. Secondly, the incorporation of fault severity data inside the state representation of reinforcement learning facilitates intelligent self-recovery measures and adaptive judgements regarding charging, discharging, and grid assistance during both normal and fault situations. Third, an extensive array of driving cycles, renewable energy scenarios, electricity price variations, and converter failure situations are meticulously analyzed within the MATLAB/Simulink framework. In contrast to conventional control techniques, such assessments enhance energy efficiency, battery usage, operational dependability, and grid support functionalities.

The rest of this work is organised as follows. The system architecture and mathematical modelling of the suggested framework are presented in detail in Section II. Section III offers the entire DRL-based energy management strategy and fault tolerance control mechanisms. Section IV offers the performance evaluation and simulation findings. Finally, Section V ends the work and also recommends directions for additional investigation.

II. LITERARTURE REVIEW

The growing popularity of electric cars (EVs) and developments in vehicle-to-grid (V2G) technology have spurred significant study in intelligent energy management, fault detection, and robust control systems. Recent advances in deep reinforcement learning, digital twins, and artificial intelligence have made it possible to create flexible solutions that can meet the intricate operational needs of contemporary electric car driving systems.

To guarantee the effectiveness of charging during grid failures, Pedapati and Abhilash [6] created a strong framework for managing electric vehicle charging. Digital twin technology and reinforcement learning serve as the foundation for this system. According to research, virtual depictions of charging infrastructure can enhance system awareness and allow for more adaptable charging choices as grid conditions change. Despite the presence of a suggested technique to increase charging resilience, the study mainly concentrated on the charging station's functioning and did not look at integrated fault-tolerant control within the vehicle drive system.

Rahmati and Rahmati presented a federated learning and vehicle-to-everything (V2X) communication-based diagnostic and energy management system for hybrid electric vehicles. They used distributed intelligence to enhance energy efficiency, detect defects, and safeguard data privacy [7]. The platform has successfully combined problem-solving and energy management tasks. Nevertheless, it was restricted to hybrid car designs and ignored deep reinforcement learning-based decision-making procedures or real-time V2G power exchange.

Enhanced energy management frameworks for solar-powered electric vehicles have been devised as the integration of renewable energy into transportation systems becomes increasingly significant. A comprehensive assessment of energy management systems for solar-electric vehicles, which comprised architectures, optimisation methodologies, and control mechanisms, was conducted by Jamadar et al. [8]. The authors underscored the importance of improved control frameworks that incorporate a variety of energy sources, thereby optimising battery usage and operational efficiency. However, the investigation identified substantial challenges related to fault-tolerant operation and real-time adaptation, emphasising the need for more sophisticated learning-based control systems.

Future electric mobility systems must stress dependability. Amin et al. [9] conducted a comprehensive

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analysis of fault-tolerant control systems to be applied in modern electric vehicles. They devised adaptive recovery techniques, defect detection algorithms, and active and passive fault-tolerant control systems for electric drives and power electronic converters. The authors observed that intelligent fault-tolerant controllers can greatly increase system reliability. However, present solutions often function independently of energy management systems, which leads to inadequate coordination between energy optimisation and reliability enhancement objectives.

The rapid development of V2G technology has led to a substantial amount of research on its possible uses in intelligent energy systems. Yu et al. [10] undertook a detailed analysis into the operational implications, future obstacles, and facilitating technologies of V2G integration. Deep reinforcement learning has been offered as a possible solution to the optimisation and control of power systems in recent years. An improved deep reinforcement learning framework for frequency regulation in active distribution networks that includes energy storage units and electric vehicles was proposed by Taher et al. [11]. Their results revealed that the dynamic response and grid support capabilities were increased in comparison to traditional control systems. The study revealed the usefulness of DRL in grid-oriented applications; nevertheless, the learning framework was insufficient in defect detection and recovery capabilities.

Through the deployment of intelligent control techniques, digital twin technology has become a commonly utilised instrument for health monitoring and predictive maintenance applications. El Hadraoui et al. [12] suggested a distributed architecture that leverages digital twins to assist the intelligent diagnosis and health monitoring of autonomous electric car powertrains. The platform enables the continuous monitoring of vehicle components and improves the accuracy of fault prediction through real-time data processing.The significant importance of intelligent energy coordination in modern electric vehicles has been emphasised by recent advancements in smart charging infrastructure and battery management systems [13], [14], [15]. The benefits of artificial intelligence for V2G applications have been demonstrated through control algorithms for bidirectional converters utilising deep learning. The primary emphasis of multi-agent artificial intelligence frameworks for bi-directional energy networks is the role of intelligent energy coordination in future smart grids [16]. The utilisation of machine learning and deep learning techniques in electric vehicle (EV) control and energy management systems is increasing, as demonstrated by a bibliometric survey of artificial intelligence applications in EVs [17]. Recent research on model predictive control approaches has emphasised the necessity for reliable and efficient energy management for V2G applications [18].

The reviewed literature demonstrates substantial progress in intelligent energy management, reinforcement learning, fault diagnosis, and Vehicle-to-Grid technologies. Nevertheless, several important challenges remain unresolved. Most existing studies address energy management and fault-tolerant control as independent optimization problems, resulting in limited coordination between operational efficiency and system reliability. Although reinforcement learning has shown considerable potential for adaptive energy scheduling, existing approaches rarely incorporate fault severity information into the decision-making process or simultaneously consider renewable energy variability, electricity price fluctuations, battery health, and converter operating conditions. Likewise, many fault diagnosis methods focus primarily on fault detection without enabling adaptive energy management or intelligent recovery strategies during abnormal operation. These limitations indicate the need for an integrated control framework capable of jointly optimizing energy utilization, grid support, battery protection, and fault resilience under dynamically changing operating conditions. Motivated by these research gaps, the proposed Deep Reinforcement Learning-based framework combines intelligent energy management with adaptive fault-tolerant control to improve both operational efficiency and system reliability in Vehicle-to-Grid enabled electric vehicle drive systems.

III. MATHEMATICAL MODELLING OF THE PROPOSED DEEP REINFORCEMENT LEARNING-BASED ENERGY MANAGEMENT AND FAULT-TOLERANT CONTROL FRAMEWORK

This research presents a Deep Reinforcement Learning (DRL)-based framework for energy management and fault-tolerant control in vehicle-to-grid (V2G) enabled electric vehicle drive systems. The proposed architecture integrates intelligent energy management, bidirectional power flow regulation, and adaptive fault recovery algorithms to enhance battery utilisation, energy efficiency, and operational reliability under diverse driving and grid conditions. The proposed methodology is organized into three primary tiers. The first tier focuses on physical energy conversion, where energy from multiple renewable sources is converted into a usable electrical form. The second tier consists of a deep reinforcement learning-based energy management system that optimizes energy distribution, load balancing, and operational efficiency under varying conditions. The third tier incorporates a fault-tolerant control and self-recovery mechanism, enabling the system to detect faults, maintain stable operation, and restore normal functionality with minimal interruption.

The Figure 1 illustrates the overarching structure of the proposed framework. The system comprises a lithium-ion battery pack, a bidirectional DC-DC converter, a voltage source inverter, a permanent magnet synchronous motor (PMSM), a utility grid interface, renewable energy sources, and a demand response load (DRL) controller. The DRL agent continuously acquires data on battery state of charge, load demand, grid power demand, electricity pricing signals, and converter operational characteristics to ascertain the optimal control action.

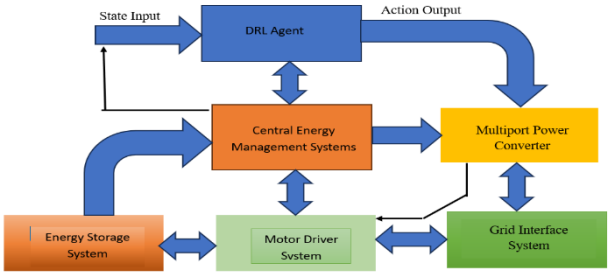


Fig. 1 Proposed Deep Reinforcement Learning-Based Energy Management and Fault-Tolerant V2G Framework for Electric Vehicle Drive Systems.

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A. Mathematical Formulation of Energy Management

The total system power balance can be represented as

$$P_{PV}(t) + P_{Grid}(t) + P_{Bat}(t) = P_{Load}(t) + P_{Loss}(t) \quad (1)$$

Where $P_{PV}(t)$ denotes renewable power generation, $P_{Grid}(t)$ represents grid power exchange, $P_{Bat}(t)$ indicates battery charging/discharging power, $P_{Load}(t)$ corresponds to EV traction demand, and $P_{Loss}(t)$represents converter and drive losses.

The battery state-of-charge evolves according to

$$SOC(t+1) = SOC(t) - \frac{\eta_b P_{Bat}(t)\Delta t}{E_{Bat}} \quad (2)$$

where η_b is battery efficiency, E_{Bat} is battery energy capacity, and Δt denotes sampling interval.

B. Deep Reinforcement Learning-Based Energy Management

The DRL agent collects information about multiple system states, including battery charge, power price signals, renewable energy availability, vehicle load demand, converter efficiency, and defect severity data. A feed-forward neural network is employed in the suggested Deep Q-Network. The two fully interconnected concealed layers consist of 128 and 64 neurons, respectively, whereas the input layer comprises the system state variables. The accessible actions are represented through the linear output layer. While the hidden layers use Rectified Linear Unit activation for superior non-linear feature learning, the output layer calculates the action-value function for every control decision. We employ the Adam optimization technique with a learning rate of 0.001 to adjust the network parameters. To increase training stability and reduce correlation between successive samples, the training is carried out using a target network, updated every 200 learning iterations, and experienced replay with a memory capacity of 10,000 transitions.

The state vector is defined as

$$S_t = \left[SOC_t, P_{PV,t}, P_{Load,t}, P_{Grid,t}, F_t\right] \quad (3)$$

where $F_t$represents the fault severity index.

The action vector is expressed as

$$A_t = \left[P_{Charge}, P_{Discharge}, M_{Conv}, M_{Grid}\right] \quad (4)$$

Where M_{Conv} denotes converter operating mode, and M_{Grid}represents grid interaction mode.

The cumulative reward function is formulated as

$$R_t = \alpha\eta_{sys} - \beta C_{energy} - \gamma D_{battery} - \delta F_t \quad (5)$$

Where $\eta_{sys}$is system efficiency, C_{energy}denotes operating cost, $D_{battery}$represents battery degradation, and $F_t$is fault severity. The weighting coefficients $\alpha, \beta, \gamma, \delta$ adjust the relative importance of each objective. To optimize cumulative rewards, the DRL controller continuously interacts with the environment and learns optimal charging, discharging, grid support, and converter operation techniques. During the policy learning stage, weighting coefficients were utilized to ensure balanced optimization of energy efficiency, operational expenses, battery degradation, and fault severity. Equal weighting was employed in initial simulated trials to avert bias towards a certain objective and to ensure uniform convergence across diverse operational conditions. The DRL agent may achieve both energy management and fault-tolerant control objectives through a constant learning process facilitated by the balanced reward structure.

C. Fault Diagnosis and Fault Severity Model

Real-time measurements from the battery, converter, motor drive, and grid interface are continuously analyzed by the fault diagnosis unit.

The normalized fault severity index is calculated as

$$F_t = \frac{1}{N}\sum_{i=1}^{N} \left| \frac{x_i - x_{i,ref}}{x_{i,ref}} \right| \quad (6)$$

Where x_i represents measured system variables, $x_{i,ref}$ denotes nominal operating values, and $N$is the total number of monitored parameters. The fault diagnosis module identifies battery deficits, converter switch failures, sensor irregularities, and motor winding issues. Depending on the severity of the problem, the controller selects corrective steps such as converter reconfiguration, power derating, battery current restriction, and emergency grid support modes. Four types of problems are identified by the fault diagnosis module: sensor malfunctions, battery abnormalities, converter switching faults, and motor drive defects. MATLAB/Simulink simulations are used to create training and evaluation datasets for different operational scenarios. Models are trained and their performance is evaluated using a variety of data sets. This methodology ensures that the stated detection efficacy is consistent with previously unidentified operational settings and allows for an objective evaluation of fault diagnosis proficiency.

D. Objective Function

The overall objective function minimized by the proposed controller is expressed as

$$J = \sum_{t=1}^{T}\left(C_{energy} + C_{degradation} + C_{fault}\right) \quad (7)$$

Where C_{energy} is the energy consumption cost,$C_{degradation}$ is the battery aging cost, and C_{fault} represents fault-induced operational penalties. To improve energy efficiency, minimise battery deterioration, and guarantee stable operation during fault circumstances, the suggested DRL-assisted self-recovery system simultaneously assesses energy optimization and fault mitigation requirements.

IV. SIMULATION RESULTS AND DISCUSSIONS

The developed Deep Reinforcement Learning (DRL)-Based Energy Management and Fault-Tolerant Control system was evaluated in MATLAB/Simulink R2024b across multiple operational situations as shown in Figure 2. The simulation model comprises a 60-kWh lithium-ion battery cell, a Vehicle-to-Grid (V2G) interface, a Permanent Magnet Synchronous Motor (PMSM) drive, a voltage source inverter, and a bidirectional DC-DC converter. The Deep Q-Network (DQN)-based deep reinforcement learning agent was trained across many scenarios, including converter breakdowns, variations in electricity costs, and swings in renewable energy output. The proposed architecture was evaluated for energy efficiency, battery utilisation, grid support capabilities, fault detection accuracy, and recovery performance. Figure 3 illustrates the

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convergence efficacy of the proposed DRL agent throughout the training phase. In the exploration phase, the cumulative reward can fluctuate significantly due to the stochastic selection of actions and the investigation of the state-action space.

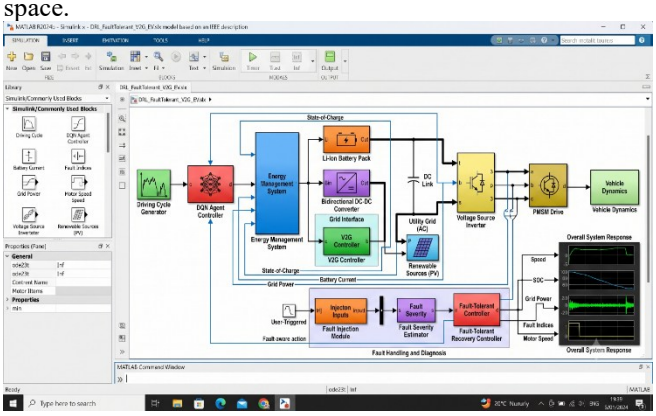


Fig. 2 Simulation of Proposed Deep Reinforcement Learning-Based Energy Management and Fault-Tolerant V2G Framework for Electric Vehicle Drive Systems.

As the quantity of training episodes increases, the cumulative incentive steadily ascends, reaching its zenith at approximately 760 episodes. The approximately $485 cumulative reward indicates that the learning process has converged effectively. The consistent reward profile indicates that the proposed DRL agent effectively acquires the appropriate charging, discharging, and fault recovery behaviours across various operating situations. The convergence characteristics of the DQN algorithm illustrate its efficacy in adaptive energy management applications.

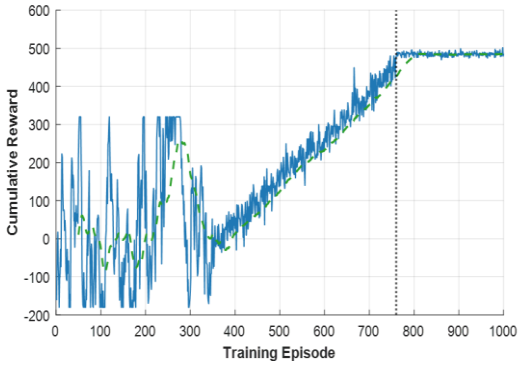


Fig. 3 Convergence characteristics of the proposed DRL agent during training.

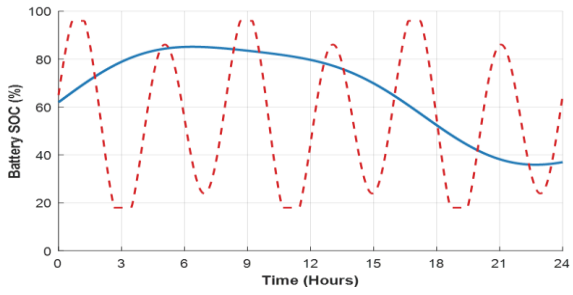


Fig. 4 Battery State-of-Charge (SOC) profile under conventional and proposed energy management strategies.

Figure 4 illustrates the State-of-Charge (SOC) profiles of the battery resulting from the proposed DRL-based controller versus the conventional rule-based controller. The standard controller demonstrates numerous oscillations in charging and discharging, leading to increased battery stress and significant variations in state of charge (SOC). The proposed controller, conversely, yields more fluid SOC trajectories and averts deep discharge scenarios. The proposed controller functions within a state of charge (SOC) range of 35% to 85%, resulting in enhanced battery health and prolonged battery lifespan. The DRL controller's ability to optimise battery utilisation while reducing degradation is emphasised by its superior charging and discharging efficiency. Figure 5 illustrates the characteristics of vehicle-to-grid power exchange. Positive grid power measurements indicate that the battery is charging, while negative readings signify that power is being sent to the utility power system. The proposed DRL controller efficiently delivers electricity to the grid during peak demand and orchestrates charging operations during periods of low cost. This adaptive scheduling method enhances grid support capabilities while reducing operating expenses.

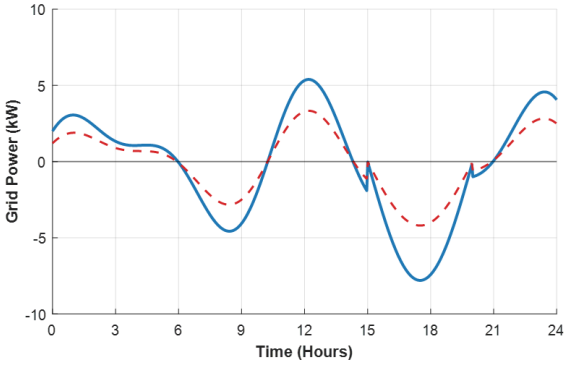


Fig. 5 Grid power exchange profile during Vehicle-to-Grid operation.

The proposed framework's maximum grid support capacity is 7.8 kW, representing an 85.7% enhancement compared to the standard controller's 4.2 kW.

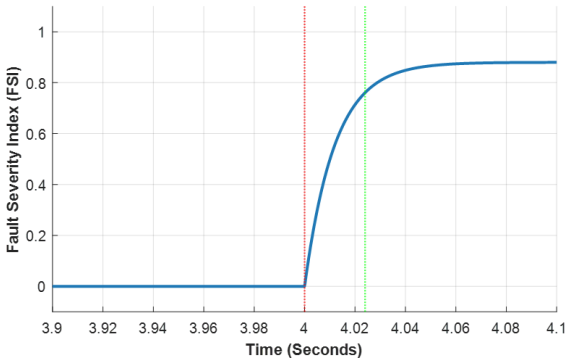


Fig. 6 Converter fault detection performance under open-switch fault conditions.

Figure 6 illustrates the fault severity index in reaction to an open switch fault event that transpired at 4 seconds. The fault severity index, which escalates rapidly following the defect's emergence, indicates abnormal operating conditions. The fault diagnosis module identifies the issue with a detection latency of merely 24 ms at 4.024 s. This allows the controller to promptly identify the issue and implement corrective actions before performance markedly declines. The collected

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data demonstrates the efficacy of the proposed problem diagnostic methodology for real-time monitoring and protection of the V2G-enabled electric vehicle drive system. Figure 7 illustrates the motor speed responses in instances of converter malfunctions. After the malfunction, there are significant discrepancies in velocity and a marked reduction in motor speed. This is the typical outcome of the controller. The proposed DRL-assisted fault recovery approach rapidly restores the PMSM speed to its reference of 3000 rpm with minimal overshoot.

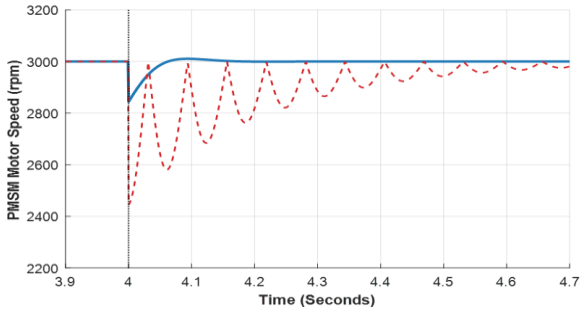


Fig. 7 Motor speed response before and after fault recovery.

The maximum speed reduction was decreased from 18.5% to 5.2%, and the recovery period has been reduced from 0.65 seconds to 0.14 seconds. The findings illustrate the capability of the proposed fault-tolerant control approach to maintain dependable motor performance in atypical conditions. Figure 8 illustrates the overall efficacy of the system in urban, highway, and V2G environments. The proposed DRL design perpetually enhances efficiency via intelligent energy scheduling and converter optimisation.

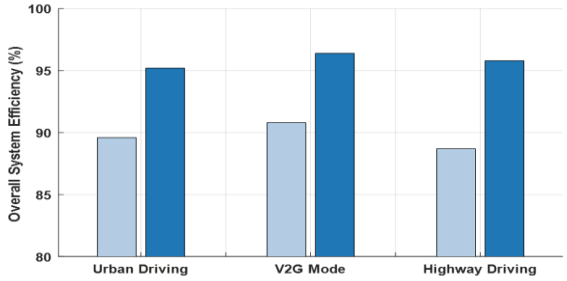


Fig. 8 System efficiency comparison under different operating conditions.

Highway driving increases from 90.8% to 96.4%, while urban driving rises from 89.6% to 95.2%. In V2G operation, the proposed controller exhibits an efficiency of 95.8%, whereas the conventional controller demonstrates an efficiency of 88.7%. The efficiency improvements indicate that the proposed system can simultaneously attain objectives in energy management and fault mitigation. Furthermore, Figure 9 illustrates the comparison of energy management systems under diverse operational settings. The proposed DRL-based controller exceeded the typical energy management method regarding overall system efficiency. The enhanced energy efficiency and superior V2G performance resulted from the strategic charging and discharging actions. The proposed controller achieved an efficiency of 95.8% in V2G operation, whereas the conventional method attained 88.7%. This was the most significant enhancement.

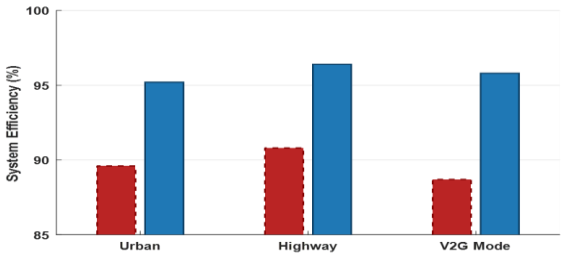


Fig. 9 Energy management systems under diverse operational settings

TABLE I. PERFORMANCE COMPARISON BETWEEN CONVENTIONAL AND PROPOSED METHOD

Performance Metric	Conventional Method	Proposed Method
Energy Efficiency	89.6–90.8%	95.2–96.4%
Battery Utilization	Lower	94.2%
Grid Support Capacity	4.2 kW	7.8 kW
Fault Detection Accuracy	Not Available	98.8%
Fault Detection Time	Higher	24 ms
Recovery Time	0.65 s	0.14 s

Table I illustrates that across all performance metrics, the proposed Deep Reinforcement Learning framework surpasses the prior rule-based approach. Enhanced battery efficiency, accelerated fault response, heightened operational economy, and superior vehicle-to-grid assistance capabilities stem from the synergistic optimization of adaptive fault recovery and energy management.

The simulation results indicate that the Deep Reinforcement Learning-based energy management and fault-tolerant control system improves operational dependability, battery utilisation, grid support capabilities, and energy efficiency. The integration of fault diagnostics and adaptive recovery techniques, which facilitate continuous operation under abnormal conditions, enhances the robustness of V2G-enabled electric vehicle drive systems. The results indicate that the proposed framework is an astute and pragmatic solution for energy management applications in next-generation electric vehicles.

V. CONCLUSION

This paper presented a framework of energy management and fault-tolerant control for vehicle-to-grid enabled electric vehicle drive systems. Deep Reinforcement Learning was employed to address issues such as varying operating conditions, unpredictability of renewable energy, changes in electricity pricing and component failure. The suggested framework incorporates intelligent energy scheduling, adaptive fault diagnostics and self-recovery control in a unified Deep Q-Network architecture that allows the coordinated improvement of battery consumption, grid support capacity, operational dependability and energy efficiency. The simulation experiments performed in the MATLAB/Simulink environment demonstrated that the proposed controller could adapt to the changing operational conditions by selecting appropriate charging, discharging and grid-support strategies while keeping the system performance stable in the presence of abnormal operating conditions. The proposed approach performed better than the conventional

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rule-based approach in terms of energy efficiency, battery utilisation, grid support capability, fault detection speed, and recovery time. The results demonstrate the effectiveness of combining reinforcement learning with adaptive fault-tolerant control. The decision-making architecture incorporating fault severity information enhances the resilience of the controller by means of fast remedial measures for faults in converter, battery, sensor and motor drive. The results suggest that the proposed method is a pragmatic and smart choice for enhancing the reliability and operational efficiency of the future vehicle-to-grid systems. The next steps will involve empirical validation on real-time hardware systems, hardware-in-the-loop execution, extensive Vehicle-to-Grid collaboration with many electric vehicles and the incorporation of sophisticated distributed learning methodologies to improve scalability, adaptability and practical application in intelligent transportation and smart grid scenarios.

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